

Fixed Income Securities
Bond features
Revision Questions for Lecture 1

After going through the material for Bond features (Yield-to-maturity), answer the following questions. Some of the points will make more sense when we cover bond pricing relationships

Notes:

- As you go through these questions, keep in mind that there is an inverse relationship between bond prices and yield to maturity (yields) – we will cover this in detail in the second week (or let me simply it and say there is negative relationship between the price you pay and the return you generate).
- The repo rate is an overnight rate so it has a huge effect on short term bonds relative to longer term bonds.

QUESTION 1 (Understanding the relationship between interest rates and bond yields)

Why do bond yields, in general, respond to interest rate changes or more specifically, to repo rate changes? [3]

Answer to Question 1

Bonds are interest bearing instruments so they are perfect substitutes to other interest bearing instruments and to a trader who prefers exposure to interest rates. So bonds respond interest changes [1]. A change in the repo rate can make bonds more attractive or less attractive than other instruments that are benchmarked against the repo rate e.g. the prime lending rate and other instruments benchmarked against the repo rate. [2]

QUESTION 2 (Surprise repo rate hike)

What is likely to happen to short term bond yields if the South African Reserve Bank (SARB) makes a surprise move to increase the repo rate by 100 basis points, explain your answer?

Answer to Question 2 (Surprise repo rate hike)

A surprise repo rate increase is likely to result in short term yields increasing [1]. (NB: if it is not a surprise we may not be able to see the reaction, the impact might have been incorporated into the yields prior to the announcement). A rate increase means related interest rates, some of which benchmarked to the repo rate, like the prime rate will also increase – this makes bonds less attractive or less competitive compared to other interest rate related instruments. This decreases the demand for short terms bonds, hence their prices will decrease. As the bond prices decreases, the yields will be increasing – so a rate cut is likely to result in bond yields increasing. [2].

QUESTION 3 (an anticipated and priced-in repo rate hike)

How does the bond market price-in (or incorporate) an anticipated rate hike on bond yields? Keep in mind that when the market price-in an anticipated rate hike we see bond yields increasing a few days prior to the announcement of the repo rate hike . Here is an example from the article I presented in class “The yields on the R186 have already spiked 10 basis points since Monday morning, as the potential hike is priced in, and could rise further if rates go up”.

Question: How does this happen? discuss what happens in reality for this to happen.

Answer to Question 3 (an anticipated and priced-in repo rate hike)

When investors are anticipating a rate hike they position themselves by either selling short term bonds or avoiding to buy short term bonds. The reason why they position themselves this way for two reasons (i) they are hoping to buy short term bonds at a lower price after the repo rate hike. So they want have cash, which they can use to buy cheaper bonds. The investors know that when the repo rate increases short term bond yields will increase as well – when short term bond yields increase their prices will decrease. In simple terms short term bonds become cheaper after the repo rate hike. (ii) bond traders will sell short term bonds to avoid capital losses, they know that when interest rates increase bond yields will increase as well while prices drop, which will result in capital losses

Selling short term bonds: In anticipation of the repo rate hike, investors who were holding short term bonds will start selling the bonds. The idea is for them to sell and hold cash so that when interest rates eventually increase, they will invest that cash in bonds with a higher yield or return and lower price or they can deposit the money in a bank and earn a higher interest after the repo rate hike (since the change in the repo rate affects both lending and deposit rates as well). As the investors sell the short-term bonds – the bond prices will be dropping (because of more supply or more sellers), and the yields will be increasing – that is how we end up observing increases in bond yields a few days prior to the rate hike announcement.

Waiting on the sidelines: Those investors who wanted to buy short term bonds will stop buying, and they will wait for interest rates to increase. The reason why they are waiting for interest rates to increase is because they are hoping that rates will increase so that they can invest their money in bonds with a higher yield, or in higher return instruments or they will buy the bonds at a cheaper price. Once buyers freeze (low demand)– the bond prices will start dropping – because there is less demand, they are no buyers – as the price is dropping the yield is increasing – that is how we end up observing increases in bond yields a few days prior to the rate hike announcement if it was well anticipated.

Recent hard evidence from US

<https://www.cnbc.com/2022/07/25/us-treasury-yields-up-ahead-of-fed-meeting-recession-risk-in-focus.html>

<https://www.cnbc.com/2022/07/27/fed-decision-july-2022-.html>

QUESTION 4

On the 18th of July 2019, the Reserve Bank of South Africa reduced the repo rate from 6.75% to 6.5%. Bond yields (for both short and long term bonds) did not experience a significant change in response to this rate cut – an indication that it was largely anticipated by market participants and it was already priced or incorporated into bond yields.

Required: Explain in detail how the expectation or anticipation of a rate cut was priced into bond yields in the days leading up the rate cut decision.[3]

Answer to question 4

When investors are anticipating a rate cut they position themselves by buying bonds, hence there will be a huge demand for bonds – by doing this, they would be hoping to profit from the bond price appreciation when repo rate is cut (bond prices increase when interest rates decrease). The high demand for bonds will push the bond prices up while pushing the yields down – hence the yields will drop before repo rate cut announcement if the market anticipates it.

QUESTION 5

What is likely to happen to short term bond yields if the South African Reserve Bank (SARB) makes a surprise move to reduce the repo rate by 100 basis points (1%), explain your answer? [3]

No answer for this question but you can figure it out by reading the answer to Question 2

Question with no answers – listen to the lecture presentation

- 1) Define the repo rate and the prime rate. What is the connection between these two interest rates?
- 2) What is the current repo rate and prime rate as of today?